

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

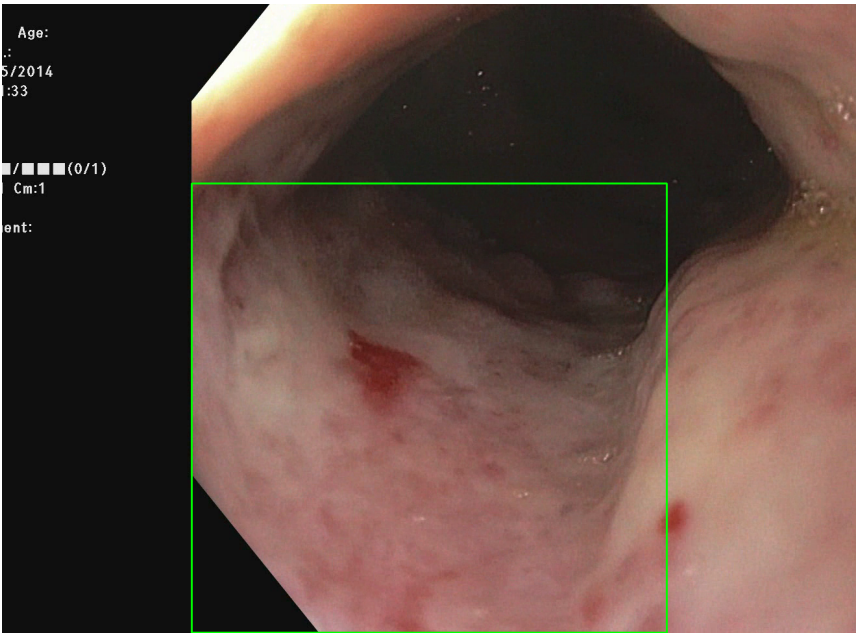
Video ID:	3e3a7ac0-4244-46cc-89a1-44ce84dd1ccf
Total Frames:	1290
Frame Rate:	25.0 fps (assumed)
Duration:	51.6 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	253	✓
RT-DETR Detections	439	✓
MedSAM2 Detections	485	✓
Consensus (3-Model Agreement)	3	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	3	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — NORMAL_MUCOSA [BASELINE] (Tracked across 25 frames)

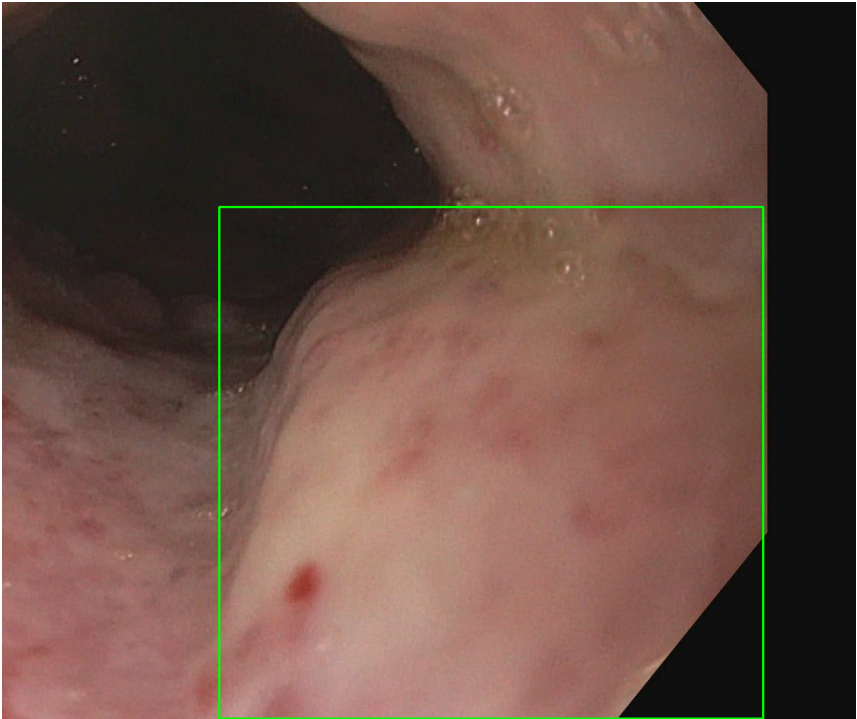


Feature	Value	Interpretation
Frame Index	351	Frame 351 of 1290
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	NORMAL_MUCOSA	Type confidence: 68.7%
Redness Score	0.118	Color-based vascularization
Relative Radius	33.5%	Polyp size as % of frame diagonal
Texture Score	0.023	Surface morphology
Vessel Visibility	0.116	Vascularization indicator
Risk Tier	BASELINE	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	67.1%	YOLO / RT-DETR / track confidence

Type Assessment: Normal mucosal pattern — no polyp morphology identified.

Insufficient evidence for a stable risk prediction - Further evaluation recommended

POLYP #2 — NORMAL_MUCOSA [BASELINE] (Tracked across 25 frames)



Feature	Value	Interpretation
Frame Index	351	Frame 351 of 1290
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 unavailable for this video)
POLYP TYPE	NORMAL_MUCOSA	Type confidence: 68.7%
Redness Score	0.118	Color-based vascularization
Relative Radius	33.5%	Polyp size as % of frame diagonal
Texture Score	0.023	Surface morphology
Vessel Visibility	0.116	Vascularization indicator
Risk Tier	BASELINE	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket

Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	62.5%	YOLO / RT-DETR / track confidence

Type Assessment: Normal mucosal pattern — no polyp morphology identified.

Insufficient evidence for a stable risk prediction - Further evaluation recommended

POLYP #3 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 23 frames)



Feature	Value	Interpretation
Frame Index	1274	Frame 1274 of 1290
Detection Method	Detector Consensus (YOLO / RT-DETR / track confidence)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 92.0%
Redness Score	0.173	Color-based vascularization

Relative Radius	20.6%	Polyp size as % of frame diagonal
Texture Score	0.063	Surface morphology
Vessel Visibility	0.737	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	53.9%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:

A total of 3 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.